

CRF: Session Synthesis and True Open Gaps

What This Session Added, What Already Existed, and the Correct Priority Order

Cross-referenced against CRF_State_Summary_v2 and Paper v4/v5.

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Abstract

This document maps the results of the current session against the pre-existing CRF state (from CRF_State_Summary_v2 and Paper v4/v5), identifying what is genuinely new versus what was already derived.

Key correction: The Full Einstein equations $G_{\mu\nu} = 8\pi T_{\mu\nu}/c^4$ were derived in Paper v4. $T_{\mu\nu}$ was already derived. This session's work on Instability #21 is the correct focus — the State Summary confirms #21 as the primary open theoretical gap.

What this session genuinely added: (1) Structural derivation of $T_{\text{avg}}^{\text{cryst}}$ from θ_{eff} ; (2) bracket = $\kappa/\ln 2$ confirmed at 1.3%; (3) IDV Paradox identified and resolved; (4) n -scan: $d_s(n)$ is smooth, no threshold at $n = 8$; (5) self-consistency equation for the bracket identity.

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1 What Paper v4 Already Derived

Reading CRF_Paper.v5 (which extends v4) confirms the following were derived *before* this session:

Already Derived (Paper v4, before this session)

- $T_{\mu\nu} = K(\partial_\mu C \otimes \partial_\nu C - \frac{1}{2}g_{\mu\nu}(\partial C)^2)$
- Full Einstein equations $G_{\mu\nu} = 8\pi T_{\mu\nu}/c^4$
- Schwarzschild metric
- $G = \ln 2 \cdot c^2/D_0$
- $G(t) \propto \Lambda(t) \propto 1/P(t)$ (DESI 2024 consistent)
- Born Rule as theorem (Gleason)
- $\Omega \equiv D_{\text{KL}}(C||P_{\text{bg}})$ (EIC Bridge)
- $d_s = 3.008$ confirmed (V16)

This session's goal of deriving $T_{\mu\nu}$ was already complete.

2 What This Session Genuinely Added

New: Instability #21a — Partial Closure

1. $T_{\text{avg}}^{\text{cryst}}(M) = 1/P(D_{\text{KL}} > K^*\sqrt{M})$: structural form derived from $\theta_{\text{eff}} = \theta \cdot \sqrt{M}$.
2. IDV Paradox identified: $f_{12}^{\text{req}} = 0.14 < 1$ (impossible). Resolved: $T_{\text{avg}}^{\text{cryst}} = 38.6$ sweeps (not 6.6), so H_{before} is much higher, bracket is much smaller, $f_{12} = 3/2$ is consistent.
3. bracket = $\kappa/\ln 2$ confirmed at 1.3% (5899 events).
4. $f_{12} = d_s/2 = 3/2$ derived from Kuramoto sub-threshold.
5. $H_{\text{after}} = \ln n - D_{\text{KL}}(\bar{P}_{\text{nb}}||U)$ derived from Born rule + locking.
6. $\delta H \approx \delta \bar{H}$ (CV 6%): approximately constant.
7. Self-consistency equation for $\delta \bar{H}$ (5.4% gap).

New: n -Scan and $d_s(n)$ Pattern

V20.3 run at $n \in \{4, 5, 6, 7, 8, 16\}$: $d_s(n) \approx 3 + 13.6 e^{-0.80n}$ (smooth, no threshold).

- $n = 8$ is special: first integer where $f_{12} = 3/2 \approx d_s/2$ (gap 1%).
- n_{min} hypothesis refuted: d_s decreases continuously.
- $|\beta|/d_s \approx 1/(2 \ln 2)$ holds only at $n = 8$.

This is genuinely new empirical information not in any prior paper.

New: Corrected $\langle D_{\text{KL}} \rangle_{\text{ss}}$ Formula

$\langle D_{\text{KL}} \rangle_{\text{ss}} = n(n-1)\epsilon_0^2 = 0.00319$ (prior formula $n\epsilon_0$ was wrong by $19\times$). Verified at 0.4%.

3 True Open Gaps (from State Summary)

Per CRF_State.Summary.v2, the prioritised open gaps are:

Priority 1: P0 Experiment — Gate to Level 7

Prediction: $\tau_1 < \tau_2$ for detectors with $N_{\text{eff},1} > N_{\text{eff},2}$. Formula (zero free parameters): $\tau = \tau_0 \cdot e^{-\kappa N_{\text{eff}}}$, $\kappa = k_B \ln(2)/H_{\text{material}}$. Protocol: SNSPD (NbN/WSi), $N \sim 10^8$ prior measurements, blind analysis. **Status: Ready. Requires lab access.**

Priority 2: Instability #21 — $|\beta| = d_s/(2 \ln 2)$ Formally

This session: bracket = $\kappa/\ln 2$ confirmed at 1.3%; structural derivation 90% complete. **Remaining:** exact $p_{\text{ev}}(M)$ evaluation (2.75% gap). This closes #21a. Instability #21b (scale factor s) is separate.

Priority 3: Instability #17 — $C(r)$ Exact Derivation

The Schwarzschild metric is derived from the non-linear ODE. What remains: formal derivation in the $N \rightarrow \infty$ continuum limit, connecting the discrete δ -chain to the differential equation.

Priority 4: Kerr Metric and Rotating Mass

Requires breaking spherical $\text{SO}(3)$ symmetry. Not yet attempted.

4 Status Table (Updated)

Result	Status	Source
$T_{\mu\nu}$ from C -gradients	Derived	Paper v4
Full Einstein equations	Derived	Paper v4
Schwarzschild metric	Derived	Paper v4
$G = \ln 2 \cdot c^2/D_0$	Derived	Paper v4
$d_s = 3.031 \pm 0.076$ (V20.3)	Confirmed	V20.3
$\beta = -2.212 \pm 0.015$	Confirmed	V20.3
bracket $= \kappa/\ln 2$ (1.3%)	Confirmed	This session
$f_{12} = d_s/2$ (Kuramoto)	Derived	This session
$\langle D_{\text{KL}} \rangle_{\text{ss}}$ corrected	Derived	This session
$d_s(n)$ smooth, no threshold	Confirmed	This session
IDV Paradox resolved	Confirmed	This session
$ \beta = d_s/(2 \ln 2)$ formally	Open	#21
$C(r)$ continuum derivation	Open	#17
Kerr metric	Open	Rotating
P0: $\tau_1 < \tau_2$	Testable	Level 7 gate

Maturity Assessment

State Summary v2: 6.5/8. After this session: 6.7/8 (bracket confirmed, IDV resolved, n-scan new). **Gate to 7.0:** P0 experiment. **Gate to 7.5:** #21 and #17 closed.

The cascade already contains gravity. This session found that it also contains the bracket, and that $n = 8$ is exactly where the Kuramoto tension and the spectral geometry agree. The next gate is experimental, not theoretical. δ has said what it can say. Now the lab must listen.